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00. Lab Setup

Purpose

This section documents the design, construction, and validation of my Red Team lab environment.

The goal is to build a realistic but manageable environment that allows me to practise offensive security techniques in a controlled setting while documenting every decision, configuration, observation, and lesson learned.

Rather than simply following setup guides, this documentation serves as a learning journal. It captures my reasoning, assumptions, questions, mistakes, and the solutions I discover throughout the build process.

Lab Architecture

The lab currently consists of three virtual machines:

- **pfSense** – Firewall/Router which serves as my Internet Core.
- **Kali Linux** – Operator machine used to simulate the attacker's workstation.
- **Ubuntu Server** – Redirector that can communicate with both the operator and the internal target.
- **Windows 10** – Internal workstation that represents the target system.

The network is intentionally designed so that the Kali operator cannot directly communicate with the Windows machine. Any interaction with the Windows host must occur through the Ubuntu redirector, mimicking segmented enterprise environments.

This design also demonstrates the operational value of using a redirector. Rather than exposing the operator's infrastructure directly to the target environment, the redirector acts as an intermediary between the two. If the redirector is detected or compromised during an engagement, it can be replaced without exposing or rebuilding the operator's primary attack infrastructure.

Objectives

The objectives of this lab are to:

- Build a repeatable, scalable Red Team practice environment.
- Understand how each system is configured and why.
- Learn networking concepts used in offensive operations.
- Validate connectivity between systems before beginning offensive exercises.

- Establish a solid foundation for future topics such as Initial Access, Enumeration, Privilege Escalation, Persistence, and Command & Control.

Documentation Structure

This folder contains the following documents:

- **Network Design** – Network layout, addressing, and communication paths.
- **PfSense** – Serve as the Internet Core.
- **Kali Operator** – Installation and configuration of the operator workstation.
- **Ubuntu Redirector** – Installation and configuration of the redirector server.
- **Windows Target** – Installation and preparation of the Windows workstation.
- **Connectivity Testing** – Verification that the lab behaves as intended.

Documentation Approach

Throughout this lab, each document may include:

- Objectives
- Configuration steps
- Commands executed
- Screenshots
- Observations
- Assumptions
- Questions raised during the build
- Problems encountered
- Lessons learned

The intention is to document not only **what was done**, but also **why it was done** and **what was learned from the process**.

Status

This lab will evolve over time. As new components are introduced or existing configurations are improved, the corresponding documentation will be updated to reflect those changes while maintaining a clear record of the learning process.